

LESSON 3-7 • SESSION 1 • GUIDED NOTES

Customary vs. Metric Conversions

Learning target – I can convert a measurement between the customary and metric systems using a given rate. • 6.RP.A.3d

WORDS TO KNOW*fill in the missing words*

WORD	WHAT IT MEANS	EXAMPLE
conversion factor <i>factor de conversión</i>	A fact that ____ two units.	1 m $\approx$ 3.28 ft
approximately <i>aproximadamente</i>	About. The symbol $\approx$ means “about ____ to.”	1 kg $\approx$ 2.2 lb
reasonableness <i>razonabilidad</i>	Checking that your answer’s ____ makes sense.	bigger unit, smaller number

KEY IDEA*complete each sentence*

- Use the fact the problem gives. Many facts are rounded, so the answer is an _____.
- Changing to a smaller unit, _____. Changing to a bigger unit, _____.

WORKED EXAMPLE*fill in the blanks as we solve*A bag weighs 44 pounds. About how many kilograms is that? (1 kg $\approx$ 2.2 lb)**Step 1** The fact: 1 kg $\approx$ ____ lb.**Step 2** A kilogram is heavier than a pound, so the number gets smaller: _____.**Step 3** $44 \div 2.2 = 20$. Check: $20 \times 2.2 = 44$.**Answer** **44 lb $\approx$ ____ kg.****YOU TRY***use the example as a model***1.** Convert. (1 kg $\approx$ 2.2 lb, 1 in = 2.54 cm)**a.** 3 kg $\approx$ ____ lb**b.** 10 in = ____ cm**2.** A fun run is 5 km long. About how many miles is that? (1 mi $\approx$ 1.6 km)*Show your work.***Answer** _____ miles**SUM IT UP***in your own words*

To check that my conversion is reasonable, I ask ____.

LESSON 3-7 • SESSION 2 • GUIDED NOTES

Multi-Step System Conversions and Dosage

Learning target – I can solve a multi-step problem that requires converting units before using a rate. · 6.RP.A.3d

WORDS TO KNOW*fill in the missing words*

WORD	WHAT IT MEANS	EXAMPLE
multi-step <i>de varios pasos</i>	Do two or more operations in ____.	convert, then multiply
dose <i>dosis</i>	The ____ of medicine to give.	120 mg
per <i>por</i>	For ____ one.	4 mg per kg

KEY IDEA*complete each sentence*

- First change the given amount to the unit the ____ uses.
- Write the ____ after every step so you know which operation comes next.

WORKED EXAMPLE*fill in the blanks as we solve*

A cat weighs 11 lb. The dose is 3 mg per kg. The syrup has 5 mg in each mL. How many mL? (1 kg $\approx$ 2.2 lb)

Step 1 Pounds to kilograms: $11 \div 2.2 =$ ____ kg.

Step 2 Dose: $5 \text{ kg} \times 3 \text{ mg per kg} =$ ____ mg.

Step 3 Syrup: $15 \text{ mg} \div 5 \text{ mg per mL} =$ ____ mL.

Answer The cat gets ____ mL of syrup.

YOU TRY*use the example as a model*

1. Complete each step. (1 kg $\approx$ 2.2 lb)

a. $44 \text{ lb} \approx$ ____ kg

b. $20 \text{ kg} \times 3 \text{ mg per kg} =$ ____ mg

2. A garden bed is 4 ft by 6 ft. One bag of soil covers 8 square feet. How many bags are needed?

Show your work.

Answer _____ bags

SUM IT UP*in your own words*

In a multi-step problem, I convert first because ____.
